

AGGREGATE FUNCTIONS with WHERE, GROUP BY, HAVING, ORDER BY

2520030030

U.Rohit Sai

STEP 1: CREATE TABLE

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator: MANAGEMENT (Server Status, Client Connections, Users and Privileges, Status and System Variables, Data Export, Data Import/Restore), INSTANCE (Startup / Shutdown, Server Logs, Options File), PERFORMANCE (Dashboard, Performance Reports, Performance Schema Setup)

Query 1 SQL File 4* Week-2* SQL File 5* x

```
1 • CREATE DATABASE IF NOT EXISTS aggregate_db;
2 • USE aggregate_db;
3 • CREATE TABLE bank_transactions (
4     txn_id INT PRIMARY KEY,
5     customer_name VARCHAR(50),
6     branch_name VARCHAR(50),
7     transaction_type VARCHAR(20),
8     amount DECIMAL(10,2),
9     transaction_date DATE
10 );
```

Output: Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected

STEP 2: INSERT DATA

MySQL Workbench

Local instance MySQL80

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Query 1 SQL File 4* Week-2* SQL File 5* x

```
1 • INSERT INTO bank_transactions VALUES
2 (101,'Rohit Sai','Hyderabad','Deposit',5000,'2024-01-05'),
3 (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
4 (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
5 (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
6 (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
7 (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
8 (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
9 (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
10 (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
11 (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
```

Output: Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sta','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

1 - SUM() :

MySQL Workbench

Local instance MySQL80 x

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PERFORMANCE

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- Performance Schema Setup

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 • SELECT SUM(amount) AS Total_Amount
2 FROM bank_transactions;
3
4
```

Result Grid

Total_Amount
70500.00

Output

Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

2 - AVG() :

MySQL Workbench

Local instance MySQL80 x

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INSTANCE

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PERFORMANCE

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Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
3 • SELECT AVG(amount) AS Average_Transaction
4 FROM bank_transactions;
5
```

Result Grid

Average_Transaction
7050.000000

Result 2 x

Output

Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

3 - MAX() :

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

7 FROM bank_transactions;

8

9

Result Grid

	Highest_Transaction
▶	15000.00

Result 3 x

Output

Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	SELECT MAX(amount) AS Highest_Transaction

4 - MIN() :

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

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Administration Schemas

Information

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

7 FROM bank_transactions;

8 SELECT MIN(amount) AS Lowest_Transaction

9 FROM bank_transactions;

Result Grid

	Lowest_Transaction
▶	1000.00

Result 4 x

Output

Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

No object selected

5 - COUNT() :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 • SELECT COUNT(*) AS Total_Transactions
2 FROM bank_transactions;
3
```

Result Grid

Total_Transactions
10

Result 5 x

Output

Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned

6 - WHERE + SUM()

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 • SELECT SUM(amount) AS Total_Deposit
2 FROM bank_transactions
3 WHERE transaction_type='Deposit';
```

Result Grid

Total_Deposit
60000.00

Result 6 x

Output

Action Output

#	Time	Action	Message
✓ 1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
✓ 2	09:40:20	USE aggregate_db	0 row(s) affected
✓ 3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
✓ 4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓ 5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
✓ 10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0...	1 row(s) returned

7 - GROUP BY :

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

```
1 • SELECT branch_name,
2 SUM(amount) AS Total_Amount
3 FROM bank_transactions
4 GROUP BY branch_name;
```

Result Grid

branch_name	Total_Amount
Hyderabad	19500.00
Vijayawada	24000.00
Vizag	27000.00

Result 7 x

Output

Action Output

#	Time	Action	Message
1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
2	09:40:20	USE aggregate_db	0 row(s) affected
3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,2024-01-05), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned

8 - GROUP BY + HAVING :

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

```
1 • SELECT branch_name,
2 SUM(amount) AS Total_Amount
3 FROM bank_transactions
4 GROUP BY branch_name
5 HAVING SUM(amount) > 20000;
```

Result Grid

branch_name	Total_Amount
Vijayawada	24000.00
Vizag	27000.00

Result 8 x

Output

Action Output

#	Time	Action	Message
1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
2	09:40:20	USE aggregate_db	0 row(s) affected
3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,2024-01-05), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name HA...	2 row(s) returned

9 - GROUP BY + ORDER BY :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

```
1 SELECT branch_name,
2 SUM(amount) AS Total_Amount
3 FROM bank_transactions
4 GROUP BY branch_name
5 ORDER BY Total_Amount DESC;
```

Result Grid

branch_name	Total_Amount
Vizag	27000.00
Vijayawada	24000.00
Hyderabad	19500.00

Output

Action Output

#	Time	Action	Message
1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
2	09:40:20	USE aggregate_db	0 row(s) affected
3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0...	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name HA...	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name O...	3 row(s) returned

10 - GROUP BY + HAVING + ORDER BY :

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

```
1 SELECT branch_name,
2 COUNT(*) AS Total_Transactions
3 FROM bank_transactions
4 GROUP BY branch_name
5 HAVING COUNT(*) >= 3
6 ORDER BY Total_Transactions DESC;
```

Result Grid

branch_name	Total_Transactions
Hyderabad	4
Vijayawada	3
Vizag	3

Output

Action Output

#	Time	Action	Message
3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branc...	0 row(s) affected
4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyd...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT ...	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name L...	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name ...	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	3 row(s) returned

11 - WHERE + COUNT() :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 SELECT COUNT(*) AS Withdrawals
2 FROM bank_transactions
3 WHERE transaction_type='Withdrawal';
4
```

Result Grid

	Withdrawals
1	4

Output

Action Output

#	Time	Action	Message
1	09:40:05	CREATE DATABASE IF NOT EXISTS aggregate_db	1 row(s) affected
2	09:40:20	USE aggregate_db	0 row(s) affected
3	09:40:24	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch...	0 row(s) affected
4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyder...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, ...	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name HA...	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name OR...	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name H...	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0, 1...	1 row(s) returned

12 - WHERE + AVG() :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 SELECT AVG(amount) AS Avg_Deposit
2 FROM bank_transactions
3 WHERE transaction_type='Deposit';
```

Result Grid

	Avg_Deposit
1	10000.000000

Output

Action Output

#	Time	Action	Message
4	09:41:57	INSERT INTO bank_transactions VALUES (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'), (102,'Sita','Hyd...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT ...	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name L...	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name ...	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0,...	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned

13 - GROUP BY + MAX() :

MySQL Workbench

Local instance MySQL80

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Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 SELECT branch_name,
2 MAX(amount) AS Highest_Amount
3 FROM bank_transactions
4 GROUP BY branch_name;
```

Result Grid

branch_name	Highest_Amount
Hyderabad	9000.00
Vijayawada	12000.00
Vizag	15000.00

Result 14

Output

Action Output

#	Time	Action	Message
5	09:43:24	SELECT SUM(amount) AS Total_Amount FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0, 1000	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned

14 - GROUP BY + MIN() :

MySQL Workbench

Local instance MySQL80

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Administration Schemas

Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 SELECT branch_name,
2 MIN(amount) AS Lowest_Amount
3 FROM bank_transactions
4 GROUP BY branch_name;
```

Result Grid

branch_name	Lowest_Amount
Hyderabad	2000.00
Vijayawada	1000.00
Vizag	4000.00

Result 15

Output

Action Output

#	Time	Action	Message
6	09:44:33	SELECT AVG(amount) AS Average_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0, 1000	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
19	10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned

15 - WHERE + ORDER BY :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5* x

```
1 SELECT *
2 FROM bank_transactions
3 WHERE amount > 8000
4 ORDER BY amount DESC;
```

Result Grid

bin_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14

bank_transactions 16 x

Output

Action Output

#	Time	Action	Message
7	09:45:36	SELECT MAX(amount) AS Highest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0, 1000	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
19	10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
20	10:03:37	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned

16 - GROUP BY + COUNT() :

MySQL Workbench

Local instance MySQL80 x

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Administration Schemas

Information

No object selected

Query 1 SQL File 4* Week-2* SQL File 5* x

```
1 SELECT branch_name,
2 COUNT(*) AS Customer_Count
3 FROM bank_transactions
4 GROUP BY branch_name;
```

Result Grid

branch_name	Customer_Count
Hyderabad	4
Vijayawada	3
Vizag	3

Result 17 x

Output

Action Output

#	Time	Action	Message
8	09:46:41	SELECT MIN(amount) AS Lowest_Transaction FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
9	09:50:20	SELECT COUNT(*) AS Total_Transactions FROM bank_transactions LIMIT 0, 1000	1 row(s) returned
10	09:51:03	SELECT SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
11	09:52:04	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
12	09:53:36	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	2 row(s) returned
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0, 1000	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0, 1000	1 row(s) returned
18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
19	10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned
20	10:03:37	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
21	10:05:12	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_name LIMIT 0, 1000	3 row(s) returned

17 - HAVING + COUNT() :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Object Info Session

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 SELECT branch_name,
2 COUNT(*) AS Total_Transactions
3 FROM bank_transactions
4 GROUP BY branch_name
5 HAVING COUNT(*) > 3;
```

Result Grid

branch_name	Total_Transactions
Hyderabad	4

Result 18 x

Output

Action Output

#	Time	Action	Message
13	09:54:29	SELECT branch_name, SUM(amount) AS Total_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0...	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0...	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0...	1 row(s) returned
18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
19	10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
20	10:03:37	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
21	10:05:12	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
22	10:07:45	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	1 row(s) returned

18 - GROUP BY + AVG() :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Object Info Session

Query 1 SQL File 4* Week-2* SQL File 5*

Limit to 1000 rows

```
1 SELECT branch_name,
2 AVG(amount) AS Avg_Amount
3 FROM bank_transactions
4 GROUP BY branch_name;
```

Result Grid

branch_name	Avg_Amount
Hyderabad	4875.000000
Vijayawada	8000.000000
Vizag	9000.000000

Result 19 x

Output

Action Output

#	Time	Action	Message
14	09:56:46	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0...	1 row(s) returned
16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0...	1 row(s) returned
17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0...	1 row(s) returned
18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
19	10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
20	10:03:37	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
21	10:05:12	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
22	10:07:45	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	1 row(s) returned
23	10:08:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned

19 - WHERE + GROUP BY + HAVING :

MySQL Workbench

Local instance MySQL80 x

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Information

No object selected

Object Info Session

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

```
1 • SELECT branch_name,
2 SUM(amount) AS Total_Deposit
3 FROM bank_transactions
4 WHERE transaction_type='Deposit'
5 GROUP BY branch_name
6 HAVING SUM(amount) > 15000;
```

Result Grid

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00

Result 20 x

Output

Action Output

#	Time	Action	Message
✓ 15	09:57:24	SELECT COUNT(*) AS Withdrawals FROM bank_transactions WHERE transaction_type='Withdrawal' LIMIT 0,...	1 row(s) returned
✓ 16	09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
✓ 17	09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
✓ 18	10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
✓ 19	10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
✓ 20	10:03:37	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓ 21	10:05:12	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
✓ 22	10:07:45	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	1 row(s) returned
✓ 23	10:08:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
✓ 24	10:10:23	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='...	2 row(s) returned

20 - GROUP BY + HAVING + ORDER BY :

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Local instance MySQL80 x

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Information

No object selected

Object Info Session

Query 1 SQL File 4* Week-2* SQL File 5* x

Limit to 1000 rows

```
1 • SELECT branch_name,
2   AVG(amount) AS Avg_Amount
3   FROM bank_transactions
4   GROUP BY branch_name
5   HAVING AVG(amount) > 7000
6   ORDER BY Avg_Amount DESC;
7
```

Result Grid

	branch_name	Avg_Amount
▶	Vizag	9000.000000
	Vijayawada	8000.000000

Result 21 x

Output

Action Output

#	Time	Action	Message
✓	16 09:58:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
✓	17 09:59:24	SELECT AVG(amount) AS Avg_Deposit FROM bank_transactions WHERE transaction_type='Deposit' LIMIT 0,...	1 row(s) returned
✓	18 10:01:25	SELECT branch_name, MAX(amount) AS Highest_Amount FROM bank_transactions GROUP BY branch_name...	3 row(s) returned
✓	19 10:02:08	SELECT branch_name, MIN(amount) AS Lowest_Amount FROM bank_transactions GROUP BY branch_name ...	3 row(s) returned
✓	20 10:03:37	SELECT * FROM bank_transactions WHERE amount > 8000 ORDER BY amount DESC LIMIT 0, 1000	4 row(s) returned
✓	21 10:05:12	SELECT branch_name, COUNT(*) AS Customer_Count FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
✓	22 10:07:45	SELECT branch_name, COUNT(*) AS Total_Transactions FROM bank_transactions GROUP BY branch_name...	1 row(s) returned
✓	23 10:08:49	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_name LI...	3 row(s) returned
✓	24 10:10:23	SELECT branch_name, SUM(amount) AS Total_Deposit FROM bank_transactions WHERE transaction_type='...	2 row(s) returned
✓	25 10:11:07	SELECT branch_name, AVG(amount) AS Avg_Amount FROM bank_transactions GROUP BY branch_name H...	2 row(s) returned